

# Markov Decision Processes

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# 1 : MDP

MDP (fully observable) . “ ” Markov , (reward) (action) .

$$\underbrace{\langle S, P \rangle}_{\text{Markov Process}} \xrightarrow{+R, \gamma} \underbrace{\langle S, P, R, \gamma \rangle}_{\text{MRP}} \xrightarrow{+A} \underbrace{\langle S, A, P, R, \gamma \rangle}_{\text{MDP}}$$



MDP  $\pi$  MRP , MRP Markov Process . (evaluation) “ MRP ” .

## 2 Markov Process

### 2.1 Markov

$$\mathbb{P}[S_{t+1} | S_t] = \mathbb{P}[S_{t+1} | S_1, \dots, S_t]$$

$S_t$  (sufficient statistic) . history .

### 2.2

$$P_{ss'} = \mathbb{P}[S_{t+1} = s' | S_t = s], \quad \sum_{s'} P_{ss'} = 1 \quad ( = 1)$$

Markov Process (Chain)  $\langle S, P \rangle$  —  $S$   $P$  . .

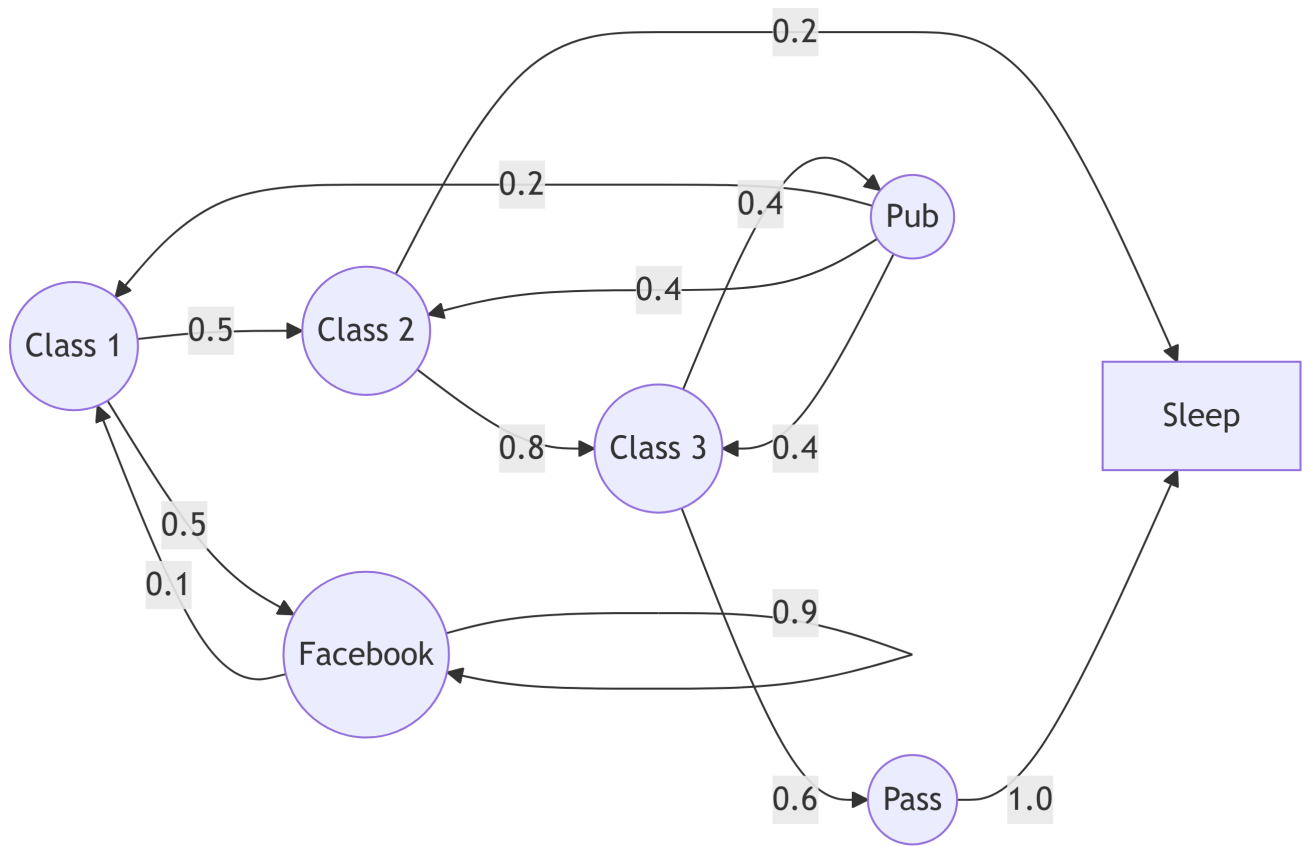
### 2.3 Student Markov Chain

(sample episode) : C1  $\rightarrow$  C2  $\rightarrow$  C3  $\rightarrow$  Pass  $\rightarrow$  Sleep, C1  $\rightarrow$  FB  $\rightarrow$  FB  $\rightarrow$  C1  $\rightarrow$  C2  $\rightarrow$  Sleep .

## 3 Markov Reward Process (MRP)

MRP =  $\langle S, P, R, \gamma \rangle$  — Markov chain (value) .

- :  $R_s = \mathbb{E}[R_{t+1} | S_t = s]$
- :  $\gamma \in [0, 1]$



1: Student Markov Chain

### 3.1 Return ( )

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

**i**

$$s \quad R_{t+1} \cdot G_t \quad , \quad v(s) \quad .$$

### 3.2 — $\gamma$

- ( )
- (cyclic) Markov process **return**
- 
- 
- /
- (terminate)  $\gamma = 1$  (undiscounted)

### 3.3 Bellman Equation

$$v(s) = \mathbb{E}[G_t \mid S_t = s]$$

Return + .

$$\begin{aligned} v(s) &= \mathbb{E}[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\ &= \mathbb{E}[R_{t+1} + \gamma v(S_{t+1}) \mid S_t = s] \\ &= R_s + \gamma \sum_{s'} P_{ss'} v(s') \end{aligned}$$

$$\mathbb{E}[G_{t+1} \mid S_{t+1} = s'] = v(s') \quad \text{tower property} \quad .$$

### 3.4

$$v = R + \gamma P v \implies (I - \gamma P) v = R \implies v = (I - \gamma P)^{-1} R$$

- ,  $O(n^3)$  ( $n = |S|$ ).
- $\gamma < 1 \implies \rho(\gamma P) < 1 \implies (I - \gamma P)^{-1}$  .  $n$  DP / Monte-Carlo / TD .

### 3.5 $\gamma$

Sleep (terminal,  $v = 0$ )  $\{C1, C2, C3, \text{Pass}, \text{Pub}, FB\}$   $v_T = (I - \gamma Q)^{-1} R_T$  .  
 $Q$  ,  $\gamma = 1 \implies \rho(Q) < 1$  .

```

import numpy as np
import matplotlib.pyplot as plt

states = ["C1", "C2", "C3", "Pass", "Pub", "FB"]
idx = {s: i for i, s in enumerate(states)}

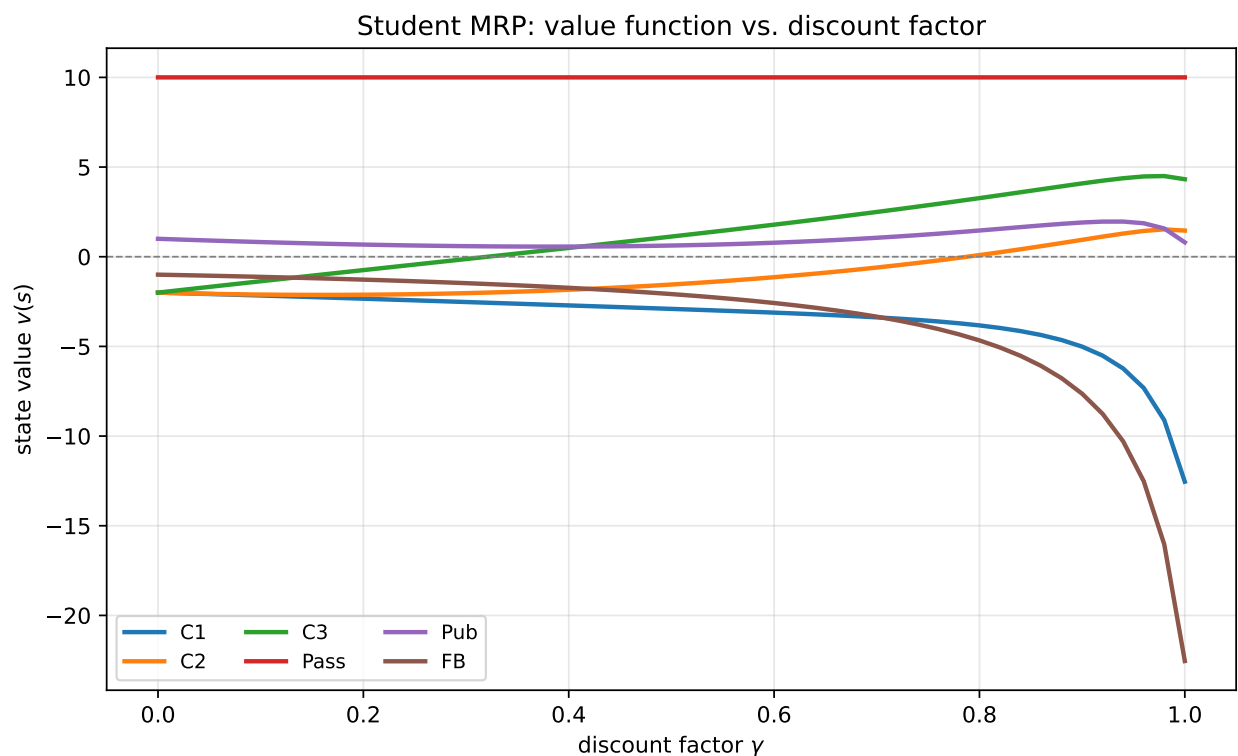
Q = np.zeros((6, 6))
Q[idx["C1"], idx["FB"]] = 0.5; Q[idx["C1"], idx["C2"]] = 0.5
Q[idx["C2"], idx["C3"]] = 0.8 # C2 -> Sleep 0.2 (terminal)
Q[idx["C3"], idx["Pub"]] = 0.4; Q[idx["C3"], idx["Pass"]] = 0.6
# Pass -> Sleep 1.0 (terminal): 0
Q[idx["Pub"], idx["C1"]] = 0.2
Q[idx["Pub"], idx["C2"]] = 0.4
Q[idx["Pub"], idx["C3"]] = 0.4
Q[idx["FB"], idx["FB"]] = 0.9; Q[idx["FB"], idx["C1"]] = 0.1

R = np.array([-2, -2, -2, 10, 1, -1], dtype=float)

gammas = np.linspace(0.0, 1.0, 51)
V = np.array([np.linalg.solve(np.eye(6) - g * Q, R) for g in gammas])

fig, ax = plt.subplots(figsize=(8, 5))
for j, s in enumerate(states):
    ax.plot(gammas, V[:, j], marker="", linewidth=2, label=s)
ax.axhline(0, color="gray", linewidth=0.8, linestyle="--")
ax.set_xlabel(r"discount factor $\gamma$")
ax.set_ylabel(r"state value $v(s)$")
ax.set_title("Student MRP: value function vs. discount factor")
ax.legend(ncol=3, fontsize=9)
ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()

```



2:    gamma            v(s)



$\gamma = 0$      $v(s) = R_s ( \quad ) \cdot \gamma$     ,     $R = +10$  Pass    C3, C2    .    0.9  
 FB     $\gamma$      $R = -1$     ( $\gamma = 1$      $v(FB) \approx -22.5$ ).

| $\gamma$ | C1     | C2    | C3    | Pass   | Pub   | FB     |
|----------|--------|-------|-------|--------|-------|--------|
| 0.0      | -2.00  | -2.00 | -2.00 | +10.00 | +1.00 | -1.00  |
| 0.5      | -2.91  | -1.55 | +1.12 | +10.00 | +0.62 | -2.08  |
| 0.9      | -5.01  | +0.94 | +4.09 | +10.00 | +1.91 | -7.64  |
| 1.0      | -12.54 | +1.46 | +4.32 | +10.00 | +0.80 | -22.54 |

## 4 Markov Decision Process (MDP)

MDP =  $\langle S, A, P, R, \gamma \rangle$  — MRP (decision) .

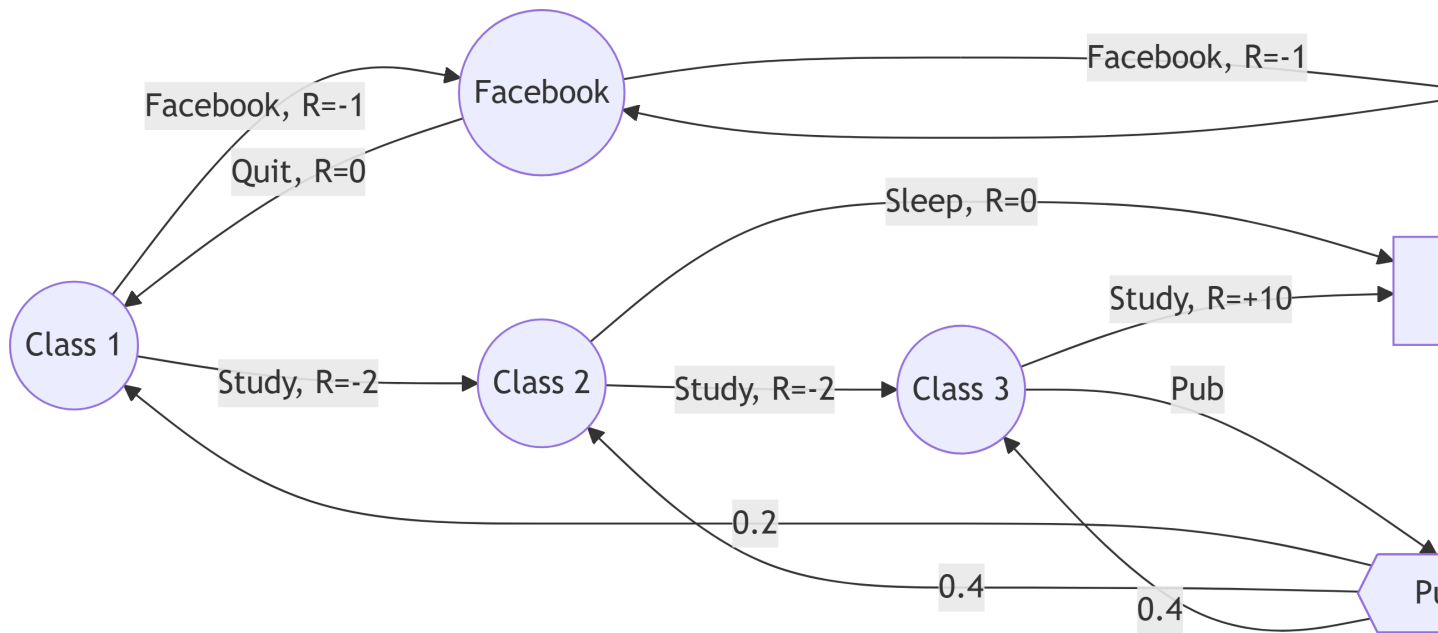
$$P_{ss'}^a = \mathbb{P}[S_{t+1} = s' \mid S_t = s, A_t = a], \quad R_s^a = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]$$

#### 4.1 (policy)

$$\pi(a | s) = \mathbb{P}[A_t = a | S_t = s]$$

(stationary), . history Markov .

#### 4.2 Student MDP



3: Student MDP ( = action / reward)

**i**

MRP “ ” , MDP “ ” . Pub (0.2/0.4/0.4) , .

#### 4.3 → MRP

$\pi$   $S_1, S_2, \dots$  Markov process  $\langle S, P^\pi \rangle$ , - MRP  $\langle S, P^\pi, R^\pi, \gamma \rangle$  .

$$P_{ss'}^\pi = \sum_a \pi(a | s) P_{ss'}^a, \quad R_s^\pi = \sum_a \pi(a | s) R_s^a$$

#### 4.4

$$v_\pi(s) = \mathbb{E}_\pi[G_t | S_t = s], \quad q_\pi(s, a) = \mathbb{E}_\pi[G_t | S_t = s, A_t = a]$$

$v_\pi$  ,  $q_\pi$  ( , ) .

## 4.5 Bellman Expectation Equation

$v$   $q$  .

$$\begin{aligned} v_{\pi}(s) &= \sum_a \pi(a | s) q_{\pi}(s, a) \\ q_{\pi}(s, a) &= R_s^a + \gamma \sum_{s'} P_{ss'}^a v_{\pi}(s') \\ v_{\pi}(s) &= \sum_a \pi(a | s) \left( R_s^a + \gamma \sum_{s'} P_{ss'}^a v_{\pi}(s') \right) \\ q_{\pi}(s, a) &= R_s^a + \gamma \sum_{s'} P_{ss'}^a \sum_{a'} \pi(a' | s') q_{\pi}(s', a') \end{aligned}$$

:

$$v_{\pi} = R^{\pi} + \gamma P^{\pi} v_{\pi} \implies v_{\pi} = (I - \gamma P^{\pi})^{-1} R^{\pi}$$

MRP .

## 5 (Optimality)

### 5.1

$$v_*(s) = \max_{\pi} v_{\pi}(s), \quad q_*(s, a) = \max_{\pi} q_{\pi}(s, a)$$

MDP “ ”  $v_*$  (  $q_*$  ) .

### 5.2

MDP .

1.  $\pi_*$  .
2.  $v_{\pi_*}(s) = v_*(s)$  .
3.  $q_{\pi_*}(s, a) = q_*(s, a)$  .

### 5.3

$q_*$  greedy (deterministic) .

$$\pi_*(a | s) = \begin{cases} 1 & \text{if } a = \arg \max_{a \in A} q_*(s, a) \\ 0 & \text{otherwise} \end{cases}$$





•  $q_*$  MDP •

## 5.4 Bellman Optimality Equation

$$\begin{aligned} v_*(s) &= \max_a q_*(s, a) \\ q_*(s, a) &= R_s^a + \gamma \sum_{s'} P_{ss'}^a v_*(s') \\ v_*(s) &= \max_a \left( R_s^a + \gamma \sum_{s'} P_{ss'}^a v_*(s') \right) \\ q_*(s, a) &= R_s^a + \gamma \sum_{s'} P_{ss'}^a \max_{a'} q_*(s', a') \end{aligned}$$

## 5.5 Expectation vs. Optimality

|                               |                             |                                 |
|-------------------------------|-----------------------------|---------------------------------|
| Bellman<br><b>Expectation</b> | $\sum_a \pi(a   s) (\cdot)$ | $(I - \gamma P^\pi)^{-1} R^\pi$ |
| Bellman<br><b>Optimality</b>  | $\max_a (\cdot)$            | $\rightarrow$                   |

max , Value Iteration, Policy Iteration, Q-learning, Sarsa •

## 6 Extensions to MDPs

### 6.1 • MDP

- (countably infinite) / :
- / : -2 (LQR)
- : → **HJB(Hamilton-Jacobi-Bellman)** , Bellman time-step  $\rightarrow 0$

### 6.2 POMDP ( )

$\langle S, A, O, P, R, Z, \gamma \rangle$ . MDP = HMM.

- **Belief state**  $b(h)$ : history  $h$
- POMDP ( ) **history tree** ( ) **belief state tree**

### 6.3 Undiscounted / MDP

- **Ergodic** Markov process = recurrent( ) + aperiodic( )
  - $\pi$  chain ergodic MDP ergodic
  - ergodic MDP  $\rho^\pi$
  - **average-reward Bellman equation** ( differential value function )
- 

## 7



1.  $v = (I - \gamma P)^{-1} R$  — Bellman . (MRP/leaky bucket . )
2. **Expectation vs. Optimality** 4 —  $v \leftrightarrow q$  . max vs.  $\sum_a \pi$  / .
3.  $\gamma$  6 .
4.  $+ q_*$  greedy .
5.  $\rightarrow$  **MRP**  $(P^\pi, R^\pi)$  — .

leaky bucket · contention “Markov chain ” , .